

Program 9:

Project Structure:

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->namespaces.yaml

->pods.yaml

Step 1: Create namespaces.yaml

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ open pods.yaml
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ cat namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$
```

Step 2: Create pods.yaml

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ cat pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
```

Step 3: Apply namespaces and pods

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl apply -f namespaces.yaml
namespace/dev unchanged
namespace/prod unchanged
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

Step 4: Verify pods in each namespace

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           11s
frontend-pod   1/1     Running   0           11s
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl get pods -n prod
NAME          READY   STATUS             RESTARTS   AGE
backend-pod    0/1     ContainerCreating   0           17s
frontend-pod   0/1     ContainerCreating   0           17s
```

Step 5 :Filter pods by label

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl get pods -n prod -l app=frontend
NAME          READY   STATUS             RESTARTS   AGE
frontend-pod   0/1     ContainerCreating   0           25s
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS             RESTARTS   AGE
backend-pod    0/1     ContainerCreating   0           30s
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl get pods -n dev -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           36s
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0           43s
```

Step 5 :Cleanup Process

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-787v6" deleted
pod "nginx-deployment-6f9664446b-vjh8w" deleted
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl delete svc --all
service "kubernetes" deleted
service "nginx-service" deleted
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program9$ kubectl delete deployments -n prod --all
```